INCH-POUND

A-A-52460A
16 May 2019
SUPERSEDING
A-A-52460
11 August 1993

# COMMERCIAL ITEM DESCRIPTION (CID)

# FILTER, FLUID, PRESSURE-AUTOMOTIVE FUEL (21 GPH FILTRATION)

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This CID covers requirements for one type of filter assembly with a coarse filtration element for use in all the fuel systems of internal combustion gasoline engines and accessories.

2. SALIENT CHARACTERISTICS.

2.1 Materials. Unless otherwise specified herein, the materials used shall be in accordance with the manufacturer's material specifications for pressure fluid filters. The use of recovered materials made in conformance with regulatory requirements is acceptable providing that all requirements of this CID are met (see note 6.5).

2.2. Cover (head). The cover shall be made of a material which shall allow the filter assembly to perform to the requirements of this CID. If made of steel, cover material shall be zinc coated and the finished cover shall be treated with a corrosion resistant inhibitor as per ASTM-D6386.

2.3 Bowl. The filter bowl material shall be brass, annealed, Unified Numbering System (UNS) C24000, C26000, or C26800 per ASTM B36/B36M; or zinc coated steel strip, carbon, cold rolled, per ASTM A109/A109M. Steel bowls shall be treated with a corrosion resistant inhibitor as per ASTM- D6386.

Comments, suggestions, or questions on this document should be sent by email to mailto: usarmy.detroit.rdecom.mbx.tardec-standardization@mail.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 2910

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2.4 Filter element. The filter element shall be made of a material which shall allow the filter assembly to perform to the requirements of this CID.

2.5 Bowl gasket. The filter bowl gasket shall be made of a rubber conforming to ASTM-D2000, classification number M5BG610A14B24E014F17 or M5BG710A14B24E014F17.

3. REGULATORY REQUIREMENTS

3.1 Filter assembly. Unless otherwise specified in figure one (1), the filter assembly and replacing element shall be constructed in accordance with the manufacturer's specifications/drawings. The filter bowl shall be compatible with the bowl gasket.

3.2 Bowl gasket. The filter bowl gasket shall be in accordance with CID A-A-52468, part number A52468-5.

3.3 Servicing. Servicing, including cleaning and reassembly shall be easily accomplished without disturbing the filter assembly connection to the engine and shall be designed to prevent improper assembly.

4. PRODUCT CONFORMANCE PROVISIONS

4.1 Pressure-temperature resistance. The strainer shall show no signs of leakage or permanent deformation after being pressure tested at -65°+ 3 degrees Fahrenheit (°F), 80 +3°F, and 160 +3°F. The pressure used in this test shall be 15 ± 1 pounds per square inch (psi). Test fluids shall be used for the -65°F and 80°F tests. Air shall be used for the 160°F test. For all tests the pressure shall be applied for 5 minutes and reduced to zero.

4.2 Flow rate. The flow rate of the fuel filter assembly shall be 21 gallons per hour (gph). The maximum differential pressure across the filter assembly shall not exceed 6 psi. Test fluid shall be a "Mineral Spirits" or "Petroleum Spirits" solvent (Stoddard solvent) and shall have a solid contaminant content of no more than 5 milligrams per liter (mg / l) and temperature of 90° +10°F.

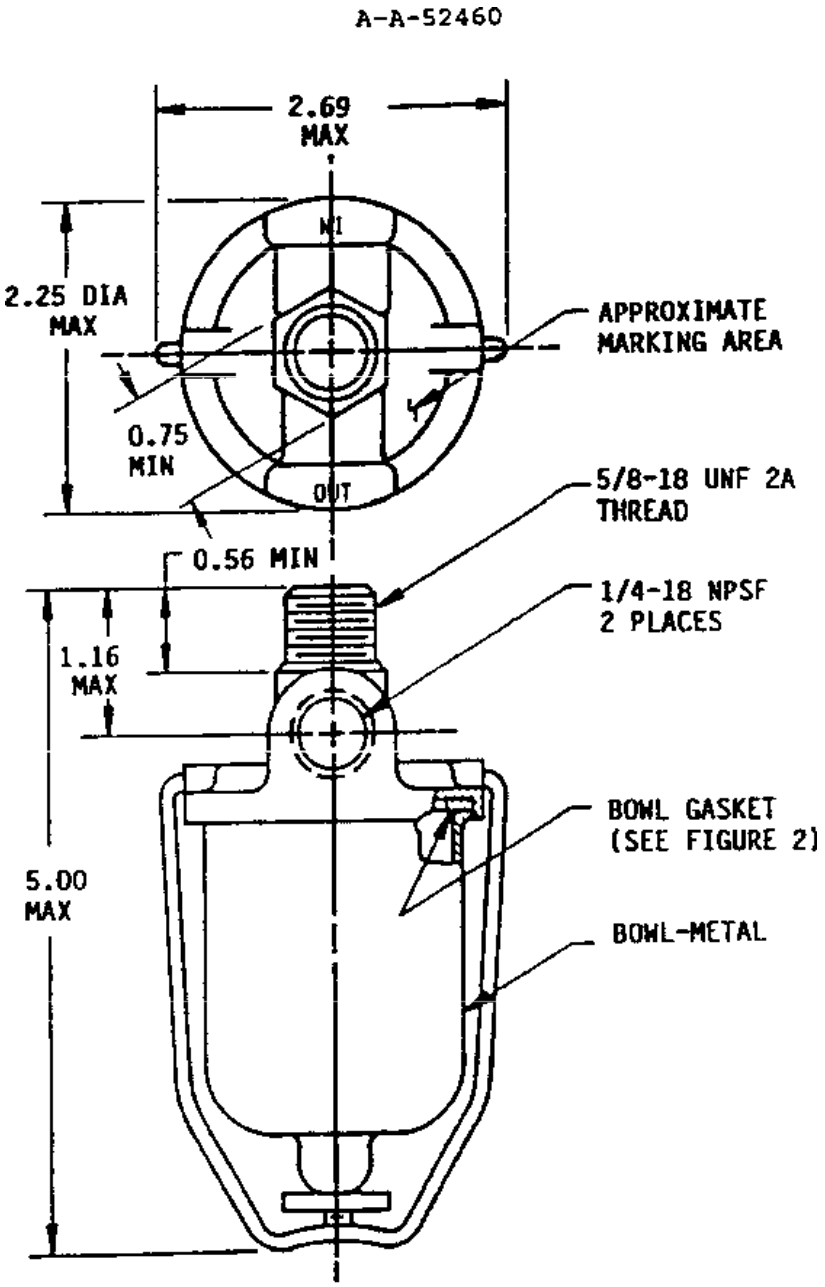
4.3 Filtering efficiency. The filtering efficiency shall conform to the requirements specified in table I. Filters shall be tested with coarse contaminants in accordance with table II. The A.C. Spark Plug Division, Coarse Air Cleaner Dust, package number 1543637 or equivalent, often referred to as "Arizona road dust" or "AC coarse dust" has been satisfactory for filter testing.

4.4 Vibration resistance. The fuel filter assembly shall have no evidence of cracking, deformation, loosening, or leakage in the body, at the gasket, or at the fittings after exposure to a sinusoidal motion for three (3) hours along each axis. A frequency range of five (5) to 500 cycles per seconds shall be used. The sweep time for the frequency range of five (5) Hertz (Hz) to 500 Hz and return to five (5) Hz shall be 15 minutes.

4.4. Identification and marking. Identification and marking shall be permanent and legible and shall include, as a minimum, the manufacturer's CAGE code and the part identification number (PIN), A52460-1 (see figure one (1) and note 6.3).



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| PIN | Former MS Part number |
|---|---|
| A52460-1 | MS51087-1 |

Notes:

1. Dimensions are in inches.
2. MAX = maximum.
3. MIN = minimum.

FIGURE 1. Filter, fluid, pressure-automotive fuel (21 gph).



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TABLE I. Filtering efficiency.

| Test duration (hours) | Flow rate (gph) | Maximum contaminant add rate-grams/hour | Minimum filtering efficiency percent (%) at 0.1 hour | Minimum filtering efficiency (%) at 1.0 hour | Minimum filtering efficiency (%) at end of test |
|---|---|---|---|---|---|
| 3 | 21 | 0.6 | 65 | 85 | 95 |

TABLE II. Contaminant Size

| Coarse contaminant | |
|---|---|
| Particle size in micrometer | Percent by weight |
| 0-5 | 12 ± 2 |
| 5-10 | 12 ± 3 |
| 10-20 | 14 ± 3 |
| 20-40 | 23 ± 3 |
| 40-80 | 30 ± 3 |
| 80-200 | 9 ± 3 |

4.5 Responsibility for inspection. The contractor is responsible for the performance of all inspections (examinations and tests).

4.6 Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this commercial item description and the product conforms to the producer's own drawings, specifications, workmanship standards, and quality assurance practices. Items with known defects shall not be submitted for Government acceptance. The Government reserves the right to require proof of such conformance prior to the first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

## 5 PACKAGING

5.1 Packaging. Preservation, packing, and marking shall be as specified in the contract or order (see note 6.3).



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6 NOTES.

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Government specifications and standards. Copies of A-A-52468 “Gasket: Filter, Fluid, Pressure, Automotive Fuel” are available online at https://quicksearch.dla.mil.

6.2 Non-Government publications. Copies of ASTM-A109/A109M “Standard Specification for Steel Strip, Carbon, (0.25 Maximum Percent), Cold Rolled”, ASTM-B36/B36M “Standard Specification for Brass Plate, Sheet, Strip, and Rolled Bar”, ASTM-D2000 “Rubber Products in Automotive Application”, and ASTM-D6386 “Preparation of Zinc (Hot-Dip Galvanized) Coated Iron and Steel Product and Hardware Surfaces for Painting” are available from www.astm.org or ASTM International, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.

6.3 Ordering data. The contract or order should specify the following:

a. Title, number, and date of this CID.
b. Type of strainer.
c. Part identification number (PIN).
d. Selection of applicable level and packaging requirements.

6.4 Cross-reference data. Fuel filters conforming to this CID are interchangeable/substitutable with fuel filters conforming to MS51087B, dated 16 February 1988 and applicable sections of A-A-52472, A-A-52461, and A-A-52468.

6.5 Regulatory requirements. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practicable.

Military Interests:

Custodians
Army - AT
Air Force – 70

Review Activities
Air Force - 170
DLA – CC

Civil Agency – Coordination Activity
GSA - FSS

Preparing Activity
Army - AT

(Project 2910-2019-003)

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